

SHARWARI AKRE

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Software Engineer | Backend Systems • Agentic AI • LLM Applications

EDUCATION

M.S. in Computer Science

North Carolina State University, Raleigh, NC

August 2024 – May 2026

GPA: 3.73 / 4.0

B.E. in Information Technology

Marathwada Mitra Mandal's College of Engineering, Pune, India

July 2019 – May 2023

GPA: 3.8 / 4.0

TECHNICAL SKILLS

Languages: Python, Java, JavaScript/TypeScript, C++, SQL

ML & Data: scikit-learn, NumPy, Pandas, Librosa, PyTorch, CodeBERT, OpenAI API, Groq, Ollama, pgvector

Backend & Frontend: Node.js, Express, Django, FastAPI, Flask, React, Next.js, REST APIs

Cloud & Databases: AWS, GCP, Docker, PostgreSQL, MongoDB, Kubernetes, Linux

Tools & Practices: Git, Postman, Agile/Scrum, OpenAPI/Swagger, Jira, JMeter, Flawfinder, Twilio

PROJECTS

Aria Companion - AI Voice Agent for Elderly Care | AWS, FastAPI, Twilio, Groq, Whisper, pgvector

[GitHub](#)

- Deployed a **proactive AI voice companion** on AWS EC2 with Elastic IP and EBS-backed storage; autonomous daily calls orchestrated via Twilio, Faster-Whisper STT, Kokoro TTS, and multi-turn LLM conversation management.
- Implemented **episodic memory** via LLM fact extraction and pgvector cosine similarity; TTS audio served from AWS S3.
- Dual-signal mood scoring (librosa acoustics + LLM sentiment) with contradiction detection; Prometheus/Grafana pipeline instrumentation resolving p95 inference bottleneck.

Meeting Negotiator - Multi-Agent Scheduling | LangGraph, FastAPI, React, Google Calendar API, Ollama

[GitHub](#)

- Delivers **~10s end-to-end** scheduling negotiation via a 4-agent LangGraph pipeline; advocate agents propose slots per participant; an arbitrator runs deterministic interval-intersection on 20+ real calendar slots before LLM selection.
- Integrated **Google Calendar freebusy API** with per-participant OAuth and automatic event creation on consensus.
- 3-round negotiation loop with escalation fallback; 45 tests across overlap logic, graph runs, and state schema validated via GitHub Actions CI.

CodeNarrator | FastAPI, Python, Ollama, D3.js

[GitHub](#)

- Developed an **autonomous codebase agent** that self-directs exploration cutting onboarding time from ~2 hours to 5 minutes.
- Integrated a local **Qwen2.5-Coder 7B model via Ollama** to generate AI architectural summaries like component breakdowns and dependency explanations with no external API or cloud cost.
- Produced self-contained HTML reports with **interactive D3.js dependency graphs** for navigating unfamiliar codebases.

WORK EXPERIENCE

Product Engineering Intern | Trust2Connect

May 2025 – July 2025

- Eliminated API timeout issues under peak load and **reduced latency by 40%** by identifying and resolving connection-per-query bottlenecks via connection pooling.
- **Achieved sub-200 ms notification latency** by designing and deploying a Pub/Sub-based communication system.
- Enabled scalable, secure document storage across 10+ microservices using GCP Storage and AlloyDB (PostgreSQL).
- Deployed application services to a **Kubernetes**-based dev environment, enabling consistent team rollouts.
- Standardized API contracts using **Apigee**-integrated REST endpoints and OpenAPI/Swagger specs; engineered **React** components to consume and surface the APIs on the frontend.

Software Developer Trainee | Capgemini

December 2023 – July 2024

- Applied indexing strategies and query optimization techniques to improve SQL and PL/SQL query performance.
- Automated **UNIX-based file operations** and logging workflows, reducing error rates and manual intervention.
- Collaborated on 5+ dashboards using **Power BI and Tableau**, enabling real-time KPI tracking across business units.

CERTIFICATIONS

AWS Certified Cloud Practitioner (CCP) — Amazon Web Services, 2026

[View Badge](#)